

RVM System — Use Case Estimation (Fibonacci / Planning Poker)

Team size: 5

Scrum approach: 5 working days per sprint

2. Main Actors & External Systems

Actor / System	Role	Key Interactions
Administrator User	Manages users, configurations, monitoring, and reporting	Manage Authentication, Manage Inventory, Manage Analytics, Manage Rewards (admin ops)
Registered User	Uses account features; earns & redeems points	Manage User Account, Manage Registered Transaction, Manage Points, Manage Rewards, Notifications
Guest User	Performs recycling transaction without account	Manage Guest Transaction, Receipts/outputs
Digital Twin System	Telemetry, optimization, monitoring for machine/inventory	Inventory signals, analytics inputs, machine status synchronization
Cafeteria System	Partner redemption channel	Reward redemption confirmation, catalog sync (if applicable)

3. Complexity Scale (Fibonacci)

Story Points	Complexity	Typical Indicators
1	Very Small	Simple CRUD, minimal rules, no integrations
2	Small	Some validation, small branching, basic persistence
3	Medium	Multiple steps, moderate rules, error handling, audit logs
5	Large	Multi-entity workflows, role checks, reporting, retries
8	Complex	Integration-heavy, concurrency/consistency needs, complex state

13	Very Complex	Hard-to-estimate scope, many edge cases, non-trivial performance/security
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4. Use Case Estimation Table

Use cases are phrased in “Manage ...” form for consistency.

UC ID & Name	Primary Actor(s)	Complexity Drivers	Story Points
UC-01 — Manage Authentication	Administrator, Registered User	Credential validation, session/token lifecycle, password reset, lockouts, audit/security events	8
UC-02 — Manage User Account	Registered User, Administrator	Profile updates, account status changes, role management, linkage to notifications/security history	5
UC-03 — Manage Registered Transaction	Registered User	End-to-end transaction workflow, item validation loop, receipt generation, points calculation, persistence	13
UC-04 — Manage Guest Transaction	Guest User	Transaction without account, reduced identity context, receipt generation, inventory + analytics updates	8
UC-05 — Manage Points	Registered User, Administrator	Ledger integrity, earn/redeem/expire, reconciliation with transactions and rewards, reporting	8
UC-06 — Manage Rewards	Registered User, Administrator	Catalog management, redemption rules, integration with Cafeteria (optional), fraud checks	5
UC-07 — Manage Inventory	Administrator, Digital Twin	Machine status + capacity thresholds, telemetry ingestion, inventory synchronization, alerts	8
UC-08 — Manage Notifications	Registered User	Preference management, event triggers (tx/points/rewards), delivery retries, templates	5

UC-09 — Manage Analytics	Administrator, Digital Twin	Aggregation, filters, export formats, dashboards, KPI definitions, performance considerations	8
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Subsystem → Use Cases

Subsystem	Use Cases
Authentication	UC-01
User Account	UC-02
Transactions	UC-03, UC-04
Points	UC-05
Rewards	UC-06
Inventory	UC-07
Notifications	UC-08
Analytics	UC-09

6. User Stories Breakdown (Low-Level)

Structure: Subsystem → Use Case → Low-level functions → User stories (As a..., I need..., so that...) + acceptance criteria.

UC-01 — Manage Authentication (8 SP)

Low-level function: Login

User story: As a Registered User, I need to log in with email and password, so that I can access my account.

1. User can log in with valid credentials and receives a session/token with expiry.
2. Invalid credentials return a safe error and record a security event.
3. Account lockout is applied after repeated failures (configurable).

Low-level function: Register

User story: As a Registered User, I need to register an account, so that I can earn points in transactions.

1. Registration enforces unique email/username.
2. System sends an OTP/verification step and confirms on success.
3. Password rules are enforced (length/complexity).

Low-level function: Reset password

User story: As a Registered User, I need to reset my password, so that I can regain access if I forget it.

1. Reset flow requires verification (OTP/email link).
2. Reset links/tokens expire and cannot be reused.
3. Audit log records password reset events.

UC-02 — Manage User Account (5 SP)

Low-level function: Update profile

User story: As a Registered User, I need to update my profile details, so that my account information stays accurate.

1. User can update allowed fields and changes persist.
2. System validates inputs (email format, required fields).
3. Updates are logged with timestamp.

Low-level function: Manage roles (admin)

User story: As an Administrator User, I need to change user roles/status, so that I can control access.

1. Admin can search/view user accounts.
2. Role/status changes apply immediately for protected endpoints.
3. All changes are audited (actor, time, before/after).

UC-03 — Manage Registered Transaction (13 SP)

Low-level function: Start/complete transaction

User story: As a Registered User, I need to start and complete a recycling transaction, so that I can receive points.

1. System creates a transaction with a clear status lifecycle.
2. System records validated items and computes totals.
3. Completing the transaction persists final totals.

Low-level function: Receipt generation

User story: As a Registered User, I need a receipt after completion, so that I have proof of the transaction.

1. Receipt is generated with totals and timestamp.
2. Receipt can be displayed/printed/downloaded.
3. Receipt references the transaction id.

UC-04 — Manage Guest Transaction (8 SP)

Low-level function: Guest transaction

User story: As a Guest User, I need to complete a transaction without an account, so that I can recycle items and receive a receipt.

1. Guest transactions can be created and completed without a user profile.
2. Receipt is generated and can be displayed/printed for the guest.
3. Transaction updates inventory counters and analytics metrics.

UC-05 — Manage Points (8 SP)

Low-level function: Points ledger

User story: As a Registered User, I need points to be recorded in a ledger, so that my balance and history are correct.

1. Points are added on transaction completion.

2. Each ledger entry references the source transaction.
3. Expired points are applied consistently and visible in history.

UC-06 — Manage Rewards (5 SP)

Low-level function: Catalog + redemption

User story: As a Registered User, I need to redeem rewards using points, so that I can exchange points for benefits.

1. Catalog shows required points and availability.
2. Redemption decreases balance and records redemption details.
3. Redemption generates a confirmation reference.

UC-07 — Manage Inventory (8 SP)

Low-level function: Inventory sync

User story: As an Administrator User, I need to view inventory per machine, so that I can plan collection/maintenance.

1. System stores per-machine inventory levels and status.
2. Inventory updates apply when transactions add items.
3. Admin can retrieve an inventory report per machine/location.

Low-level function: Threshold alerts

User story: As a System, I need to generate alerts on threshold breaches, so that issues are handled quickly.

1. Threshold rules are configurable (e.g., 80% capacity).
2. Alerts are created with severity and timestamp.
3. Alerts are visible to administrators.

UC-08 — Manage Notifications (5 SP)

Low-level function: Preferences + delivery

User story: As a Registered User, I need to configure notification preferences, so that I receive relevant updates.

1. User can enable/disable channels in preferences.
2. System sends notifications on key events (transaction complete, redemption).
3. Failures are retried or recorded with clear status.

UC-09 — Manage Analytics (8 SP)

Low-level function: Dashboard

User story: As an Administrator User, I need analytics dashboards, so that I can monitor usage and performance.

1. Admin can view key metrics over a date range.
2. Filters apply correctly and match stored data.
3. Dashboard load time is acceptable for typical ranges.

Low-level function: Export reports

User story: As an Administrator User, I need to export reports, so that I can share and archive results.

1. Admin can export CSV/PDF for a date range.
2. Export matches displayed metrics and includes headers.
3. File is downloadable and named with a timestamp.

Sprint plan (5-day sprints)

Assumption: 1 sprint targets ~13 story points of scope (planning poker velocity).

Sprint	Planned scope (user stories / use cases)	Total SP
Sprint 1	UC-01 (Auth) + start UC-02 (Account)	13
Sprint 2	Finish UC-02 + UC-07 (Inventory)	13
Sprint 3	UC-03 (Registered Transaction)	13

Sprint 4	UC-04 (Guest Transaction) + UC-05 (Points)	16
Sprint 5	UC-06 (Rewards) + UC-08 (Notifications)	10
Sprint 6	UC-09 (Analytics) + regression/integration/UAT/release	8+

Schedule (dependencies, network, deadline, Gantt + risk)

See the appended schedule section and HQ figures generated from the UC dependency graph.

Updated schedule (CPM)

Task durations and dependencies

Activity	Duration (days)	Dependencies
Start (M_START)	0	—
UC-01 Authentication (Login/Register/Reset) (T1)	10	Start (M_START)
Milestone: Authentication Ready (M1)	0	UC-01 Authentication (Login/Register/Reset) (T1)
UC-02 User Account (Profile/Roles) (T2)	10	Milestone: Authentication Ready (M1)
UC-07 Inventory (Sync/Threshold Alerts) (T3)	10	Milestone: Authentication Ready (M1)
Milestone: Core Ready (Account + Inventory) (M2)	0	UC-02 User Account (Profile/Roles) (T2), UC-07 Inventory (Sync/Threshold Alerts) (T3)
UC-03 Registered Transaction (Workflow + Receipt) (T4)	10	Milestone: Core Ready (Account + Inventory) (M2)
Milestone: Registered Transaction Ready (M3)	0	UC-03 Registered Transaction (Workflow + Receipt) (T4)
UC-04 Guest Transaction (Workflow + Receipt) (T5)	5	Milestone: Registered Transaction Ready (M3)

UC-05 Points (Ledger/Earn/Expire) (T6)	10	Milestone: Registered Transaction Ready (M3)
Milestone: Transactions + Points Ready (M4)	0	UC-04 Guest Transaction (Workflow + Receipt) (T5), UC-05 Points (Ledger/Earn/Expire) (T6)
UC-06 Rewards (Catalog + Redemption) (T7)	5	Milestone: Transactions + Points Ready (M4)
UC-08 Notifications (Preferences + Delivery) (T8)	5	UC-06 Rewards (Catalog + Redemption) (T7)
UC-09 Analytics (Dashboard) (T9)	10	Milestone: Transactions + Points Ready (M4)
UC-09 Analytics (Export Reports) (T10)	5	UC-09 Analytics (Dashboard) (T9)
Milestone: Product Features Complete (M5)	0	UC-08 Notifications (Preferences + Delivery) (T8), UC-09 Analytics (Export Reports) (T10)
System-wide Regression + Integration (T11)	5	Milestone: Product Features Complete (M5)
UAT + Release (T12)	5	System-wide Regression + Integration (T11)
Finish (M_FINISH)	0	UAT + Release (T12)

Activity network

Activity network

```mermaid

flowchart LR

%% task boxes vs milestone circles

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classDef milestone fill:#cfeef2,stroke:#1e4aa5,stroke-width:4px;

M\_START(("M\_START\nStart\n2026-01-05"))

T1["T1\nUC-01 Authentication (Login/Register/Reset)\n10 days\n2026-01-05 → 2026-01-19"]

M1(("M1\nMilestone: Authentication Ready\n2026-01-19"))

T2["T2\nUC-02 User Account (Profile/Roles)\n10 days\n2026-01-19 → 2026-02-02"]

T3["T3\nUC-07 Inventory (Sync/Threshold Alerts)\n10 days\n2026-01-19 → 2026-02-02"]

M2(("M2\nMilestone: Core Ready (Account + Inventory)\n2026-02-02"))

T4["T4\nUC-03 Registered Transaction (Workflow + Receipt)\n10 days\n2026-02-02 → 2026-02-16"]

M3(("M3\nMilestone: Registered Transaction Ready\n2026-02-16"))

T5["T5\nUC-04 Guest Transaction (Workflow + Receipt)\n5 days\n2026-02-16 → 2026-02-23"]

T6["T6\nUC-05 Points (Ledger/Earn/Expire)\n10 days\n2026-02-16 → 2026-03-02"]

M4(("M4\nMilestone: Transactions + Points Ready\n2026-03-02"))

T7["T7\nUC-06 Rewards (Catalog + Redemption)\n5 days\n2026-03-02 → 2026-03-09"]

T8["T8\nUC-08 Notifications (Preferences + Delivery)\n5 days\n2026-03-09 → 2026-03-16"]

T9["T9\nUC-09 Analytics (Dashboard)\n10 days\n2026-03-02 → 2026-03-16"]

T10["T10\nUC-09 Analytics (Export Reports)\n5 days\n2026-03-16 → 2026-03-23"]

M5(("M5\nMilestone: Product Features Complete\n2026-03-23"))

T11["T11\nSystem-wide Regression + Integration\n5 days\n2026-03-23 → 2026-03-30"]

T12["T12\nUAT + Release\n5 days\n2026-03-30 → 2026-04-06"]

M\_FINISH(("M\_FINISH\nFinish\n2026-04-06"))

M\_START -->|10d| T1

T1 --> M1

M1 -->|10d| T2

M1 -->|10d| T3

T2 --> M2

T3 --> M2

M2 -->|10d| T4

T4 --> M3

M3 -->|5d| T5

M3 -->|10d| T6

T5 --> M4

T6 --> M4

M4 -->|5d| T7

T7 -->|5d| T8

M4 -->|10d| T9

T9 -->|5d| T10

T8 --> M5

T10 --> M5

M5 -->|5d| T11

T11 -->|5d| T12

T12 --> M\_FINISH

class M\_START milestone;

class T1 task;

class M1 milestone;

class T2 task;

class T3 task;

class M2 milestone;

class T4 task;

class M3 milestone;

class T5 task;

class T6 task;

class M4 milestone;

class T7 task;

```

class T8 task;

class T9 task;

class T10 task;

class M5 milestone;

class T11 task;

class T12 task;

class M_FINISH milestone;

'''

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### Deadline and critical path

# Schedule deadline

- Total duration (critical path): **\*\*65 working days\*\***
- Critical activities (slack = 0): M\_START, T1, M1, T2, T3, M2, T4, M3, T6, M4, T9, T10, M5, T11, T12, M\_FINISH
- Start date: **\*\*2026-01-05\*\***
- Deadline (Mon–Fri): **\*\*2026-04-06\*\***

### Feature risk (baby-blue shows schedule buffer)

| Feature | Duration (days) | Slack (days) | Schedule risk |
|---------|-----------------|--------------|---------------|
|---------|-----------------|--------------|---------------|

### Activity timeline values (for Gantt)

| Activity                                         | ES | EF | LF | Slack | Critical |
|--------------------------------------------------|----|----|----|-------|----------|
| M_START — Start                                  | 0  | 0  | 0  | 0     | Yes      |
| T1 — UC-01 Authentication (Login/Register/Reset) | 0  | 10 | 10 | 0     | Yes      |
| M1 — Milestone: Authentication Ready             | 10 | 10 | 10 | 0     | Yes      |

|                                                        |    |    |    |   |     |
|--------------------------------------------------------|----|----|----|---|-----|
| T2 — UC-02 User Account (Profile/Roles)                | 10 | 20 | 20 | 0 | Yes |
| T3 — UC-07 Inventory (Sync/Threshold Alerts)           | 10 | 20 | 20 | 0 | Yes |
| M2 — Milestone: Core Ready (Account + Inventory)       | 20 | 20 | 20 | 0 | Yes |
| T4 — UC-03 Registered Transaction (Workflow + Receipt) | 20 | 30 | 30 | 0 | Yes |
| M3 — Milestone: Registered Transaction Ready           | 30 | 30 | 30 | 0 | Yes |
| T5 — UC-04 Guest Transaction (Workflow + Receipt)      | 30 | 35 | 40 | 5 | No  |
| T6 — UC-05 Points (Ledger/Earn/Expire)                 | 30 | 40 | 40 | 0 | Yes |
| M4 — Milestone: Transactions + Points Ready            | 40 | 40 | 40 | 0 | Yes |
| T7 — UC-06 Rewards (Catalog + Redemption)              | 40 | 45 | 50 | 5 | No  |
| T8 — UC-08 Notifications (Preferences + Delivery)      | 45 | 50 | 55 | 5 | No  |
| T9 — UC-09 Analytics (Dashboard)                       | 40 | 50 | 50 | 0 | Yes |
| T10 — UC-09 Analytics (Export Reports)                 | 50 | 55 | 55 | 0 | Yes |
| M5 — Milestone: Product Features Complete              | 55 | 55 | 55 | 0 | Yes |
| T11 — System-wide Regression + Integration             | 55 | 60 | 60 | 0 | Yes |
| T12 — UAT + Release                                    | 60 | 65 | 65 | 0 | Yes |
| M_FINISH — Finish                                      | 65 | 65 | 65 | 0 | Yes |

## High-quality schedule figures

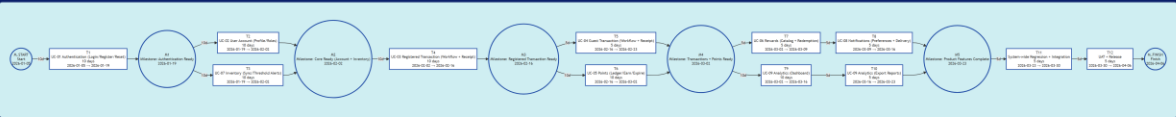
Dependency Table

Task durations and dependencies

| Activity                                               | Duration (days) | Dependencies                                                                              |
|--------------------------------------------------------|-----------------|-------------------------------------------------------------------------------------------|
| Start (M_START)                                        | 0               | —                                                                                         |
| UC-01 Authentication (Login/Register/Reset) (T1)       | 10              | Start (M_START)                                                                           |
| Milestone: Authentication Ready (M1)                   | 0               | UC-01 Authentication (Login/Register/Reset) (T1)                                          |
| UC-02 User Account (Profile/Roles) (T2)                | 10              | Milestone: Authentication Ready (M1)                                                      |
| UC-07 Inventory (Sync/Threshold Alerts) (T3)           | 10              | Milestone: Authentication Ready (M1)                                                      |
| Milestone: Core Ready (Account + Inventory) (M2)       | 0               | UC-02 User Account (Profile/Roles) (T2), UC-07 Inventory (Sync/Threshold Alerts) (T3)     |
| UC-03 Registered Transaction (Workflow + Receipt) (T4) | 10              | Milestone: Core Ready (Account + Inventory) (M2)                                          |
| Milestone: Registered Transaction Ready (M3)           | 0               | UC-03 Registered Transaction (Workflow + Receipt) (T4)                                    |
| UC-04 Guest Transaction (Workflow + Receipt) (T5)      | 5               | Milestone: Registered Transaction Ready (M3)                                              |
| UC-05 Points (Ledger/Earn/Expire) (T6)                 | 10              | Milestone: Registered Transaction Ready (M3)                                              |
| Milestone: Transactions + Points Ready (M4)            | 0               | UC-04 Guest Transaction (Workflow + Receipt) (T5), UC-05 Points (Ledger/Earn/Expire) (T6) |
| UC-06 Rewards (Catalog + Redemption) (T7)              | 5               | Milestone: Transactions + Points Ready (M4)                                               |
| UC-08 Notifications (Preferences + Delivery) (T8)      | 5               | UC-06 Rewards (Catalog + Redemption) (T7)                                                 |
| UC-09 Analytics (Dashboard) (T9)                       | 10              | Milestone: Transactions + Points Ready (M4)                                               |
| UC-09 Analytics (Export Reports) (T10)                 | 5               | UC-09 Analytics (Dashboard) (T9)                                                          |
| Milestone: Product Features Complete (M5)              | 0               | UC-08 Notifications (Preferences + Delivery) (T8), UC-09 Analytics (Export Reports) (T10) |
| System-wide Regression + Integration (T11)             | 5               | Milestone: Product Features Complete (M5)                                                 |
| UAT + Release (T12)                                    | 5               | System-wide Regression + Integration (T11)                                                |
| Finish (M_FINISH)                                      | 0               | UAT + Release (T12)                                                                       |

Activity Network

Activity network



# Activity Timeline

